

Radical expressions



1 Write the following expressions as a single radical:

a $3\sqrt{3}$

b $4\sqrt{2}$

c $2\sqrt[3]{5}$

d $7\sqrt[3]{2}$

2 Complete Statement 2 by following the example in Statement 1:

a

- Statement 1: $\sqrt{9 \cdot 11} = \sqrt{9} \cdot \sqrt{11} = 3\sqrt{11}$
- Statement 2: $\sqrt{25 \cdot 2} = \sqrt{1} \cdot \sqrt{1} = 1\sqrt{1}$

b

- Statement 1: $\sqrt{9 \cdot 4} = \sqrt{9} \cdot \sqrt{4} = 3 \cdot 2 = 6$
- Statement 2: $\sqrt{16 \cdot 25} = \sqrt{1} \cdot \sqrt{1} = 1 \cdot 1 = 1$

3 Simplify:

a $3\sqrt{81}$

b $\sqrt{49 \times 10}$

c $\sqrt{150}$

d $\frac{64}{\sqrt{64}}$

e $\frac{1}{4}\sqrt{16}$

f $\sqrt[3]{-1}$

g $\sqrt[3]{-64}$

h $\sqrt[3]{8}$

i $\sqrt[3]{8 \times 6}$

j $\sqrt[3]{81}$

k $7\sqrt[3]{125}$

l $\frac{1}{\sqrt[3]{8}}$

m $\frac{3}{2}\sqrt[3]{8}$

n $\sqrt[3]{108}$

4 Determine whether the following expressions are written in simplest form:

a $\sqrt{19}$

b $7\sqrt{50}$

c $\sqrt[3]{14}$

5 Simplify:

a $\sqrt{x^8}$

b $\sqrt[3]{m^3}$

c $\sqrt[3]{v^{18}}$

d $\sqrt[3]{6^3 x^9}$

e $\sqrt[3]{b^3 x^9}$

f $\sqrt[3]{n^9 p^9}$

g $\sqrt[3]{9^3 x^{18} y^{12}}$

h $\left(\sqrt[3]{7y^2}\right)^3$

i $\sqrt{a^2}$

j $\sqrt{b^{14}}$

k $\sqrt{4^2 x^8}$

l $\sqrt{x^8 y^{10}}$

m $\sqrt{j^2 x^4}$

n $\sqrt{k^8 x^8}$

o $\sqrt{5^2 x^{14} y^{20}}$

p $\sqrt{28a^3 y^4}$

q $\sqrt{m^4}$

r $\sqrt{64x}$

6 Simplify:



a $\frac{1}{3}\sqrt[3]{27 \times 4}$

b $\sqrt{\left(\frac{x^4}{4}\right)}$

c $\sqrt{\frac{49x^4}{64}}$

d $\sqrt{\frac{36x^{18}}{y^{20}}}$

7 Simplify:

a $\sqrt{(2x+9)^2}$

b $\sqrt{16y^2 + 24y + 9}$

c $\sqrt{x^2y^2 + 18xy^2 + 81y^2}$

8 Evaluate the following expressions:

a $\sqrt{(-3)^2 + 4^2}$

b $\sqrt{(-10)^2 - 8^2}$

c $\sqrt{(-13)^2 - (-12)^2}$

9 Evaluate the following expressions:

a $\sqrt[3]{-125} \times \sqrt[3]{27}$

b $\sqrt[3]{-64} \div \sqrt[3]{8}$

c $\sqrt[3]{-125} \times \sqrt[3]{-64}$

10 Determine whether the following statements are true or false:

a $\sqrt{(-3)^2} = -\sqrt{3^2}$

b $\sqrt{32} \neq \sqrt{-4} \times \sqrt{-8}$

c If x is a positive number, $\sqrt{x^{16}} = x^4$.

11 Assuming that $x > -3$, Simplify the following expression:

$$\sqrt{(x+3)^2}$$

12 Solve the following equation for k :

$$\sqrt[k]{y} \times \sqrt[k]{y} \times \sqrt[k]{y} = y^{\frac{1}{2}}$$